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FROM MACHINE LEARNING MODELS TO STRATEGY¹

Overview and Purpose

Many machine learning projects underperform not because the models are weak, but because their findings are not communicated in a decision-ready format. Managers must convert technical outputs into strategic narratives that align with organizational objectives, constraints, and return on investment. Prediction is only useful when it leads to concrete action. For instance, a churn prediction model should inform targeted retention offers.

The purpose of this note is to transcend technical understanding to strategic interpretation and show how to translate analytical findings into persuasive, ethical, and actionable business recommendations.

From Prediction to Insight

The key shift in using statistical analysis or machine learning for business is moving from *what* the model predicts to *why* it does so, and so *what* that means for decisions. A model's output, like a churn probability or sales forecast, is only useful as long as it is clear what is driving it and how that insight can guide better decisions.

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To do this, we use interpretation tools that show how distinct factors influence prediction. Feature importance tells us which variables matter most overall, for example, whether response time or price has a bigger effect on churn. Local explanations zoom in on individual cases, showing why the model made a specific prediction for a single customer or transaction.

For instance, imagine a churn model that predicts which customers are most likely to leave. If it shows that service response time is a stronger driver of attrition than price, it may be more effective to strengthen service response standards and ensure faster customer support performance than to offer broad discounts.

Even in a relatively simple model like logistic regression, we benefit from interpreting statistical results in plain language for communication. The model calculates the probability of an event (like churn) based on the weighted influence of different factors. Each factor has a *coefficient* (β):

- A positive coefficient means that, holding all other variables, as this variable increases, the likelihood of the event (e.g., churn) increases. A negative coefficient means the opposite; it reduces the likelihood.
- The size of the coefficient shows how strong that influence is compared with other factors, as long as variables are measured on similar scales. To ensure this, either compare variables of similar scale (i.e. Likert-Scale answers between 1 and 7) or transform them to be on the same scale (i.e. between 0 and 1).

Therefore, if ‘response time’ has a larger positive coefficient than ‘price satisfaction’, it means longer response times raise churn risk more than dissatisfaction with pricing. Furthermore, if both of them have p-values below 0.05, we know the identified relationship and churning is reliable.

Furthermore, each coefficient has a p-value, which measures the statistical significance of the variable in the model. The usual standard is that a p-value below 0.10 is considered weakly significant and a p-value below 0.05 is considered significant. Although we can leave the factual interpretation of p-value and confidence intervals for academics interested in hypothesis testing, we should remember that a lower p-value means a more statistically significant result. In plain language, a more reliable result.

Visualization for Managerial Communication

Effective visualization is essential for bridging technical and business perspectives. Core artifacts include confusion matrices (showing classification trade-offs), Receiver Operating Characteristic (ROC) curves and Area Under Curve (AUC), (for threshold-independent discrimination), lift/gain charts (highlighting business value concentration), and plots such as feature importance bars, PDP/ICE plots, and cluster visualizations for unsupervised tasks.

When presenting results, minimize jargon and pair each technical metric with a plain-language business equivalent. Highlight uncertainty through confidence intervals or variability across validation folds and explicitly connect visuals to financial or operational implications such as cost, risk, or service levels.

Model Explainability (Trust and Adoption)

The goal of explainability is to help non-technical stakeholders understand what most influences predictions and how they can act on those insights, without delving into mathematical jargon. Model explanations can be global or local. *Global* methods describe how the model behaves on average across all data, such as what features generally drive predictions, while *local* methods explain a single

prediction, showing why the model produced that specific result for one case. Together, they provide both a high-level overview and detailed case-level transparency, building trust with business stakeholders by showing that model behavior aligns with domain intuition.

Three widely applicable techniques are:

1. Feature importance (global): Highlights which variables matter most overall. This supports a “top drivers” summary—speaking to direction (“longer response time increases churn”) and relative influence (“twice as important as price”) while avoiding causal claims.
2. Effect plots or partial dependence (global): Show how predictions change as one variable varies while others stay typical, revealing trends, thresholds, and diminishing returns.
3. Per-case explanations (local, e.g., SHAP or LIME): Show how specific variables contributed to a single decision, useful for case reviews or policy checks.

When presenting, dedicate the first slide to top drivers with directional arrows and concise takeaways, the second to a single key effect plot with interpretation, and the third to two illustrative cases.

Remember that importance does not imply causality, explanations depend on current data, and fairness should always be monitored. Use explainability to decide *which lever to pull* or justify thresholds, not to prove cause and effect.

Connecting Results to Business Strategy

Interpretation should feed a continuous loop of Insight → Decision → Action → Feedback. Start by identifying who is at risk and what drivers matter. Next, decide which levers, whether pricing, service, or messaging, are feasible given constraints. Then act through targeted treatments such as offers or outreach, and finally measure uplift, recalibrate thresholds, and retrain with new data.

This approach scales across functions. For example, in marketing, cluster interpretation supports segment-specific campaigns; in finance, PD/LGD² models guide cutoff policies and capital allocation; in operations, forecasting enables inventory and staffing optimization.

Communicating to Stakeholders

Effective communication means tailoring the message to the audience. Executives want the story: The objective, the key insight, its business impact on revenue, cost, or risk, and what action to take next. Technical teams, on the other hand, need the details: data scope, modeling assumptions, validation results, monitoring plans, and known limitations. Adjust the level of technical depth accordingly.

A clear structure for executive communication is:

Objective → Method → Key Insight → Business Impact → Next Steps

This flow moves from purpose to evidence to action, helping non-technical leaders follow your logic and make informed decisions.

When presenting model decisions, explain how you chose the probability threshold (τ) that determines which cases are classified as “positive.” A model may predict probabilities (e.g., a

² PD stands for Probability of Default. LGD stands for Loss Given Default.

customer has a 0.78 chance of churning), but management must still decide where to draw the line and at what point a case warrants action.

To make that choice rational, define a simple cost function, where total expected cost is represented as $C(\tau)$.

$$C(\tau) = C_{FP} \times FP(\tau) + C_{FN} \times FN(\tau)$$

Here:

- $P(\tau)$ and $FN(\tau)$ are the number of false positives and false negatives at a given threshold.
- C_{FP} and C_{FN} are the business costs of each type of errors, such as the cost of sending an unnecessary offer versus losing a valuable customer you could have saved.³

The optimal threshold is the one that minimizes total expected cost, balancing over- and under-action according to business priorities.

When reporting results, include both the chosen threshold (τ) and the associated performance metrics (precision, recall, AUC, lift, etc.). This shows stakeholders not just how accurate the model is, but how it connects to business value and risk appetite. With an executive audience, be ready to explain metrics in plain language, while avoiding jargon.

Ethical and Strategic Responsibility

Responsible communication means clarifying data limits and uncertainty, avoiding causal claims when using correlational models, and testing for group fairness (e.g., disparate impact ratios or error parity). Privacy, consent, and data retention must comply with regulations, and systems should be monitored for drift or performance decay.

An ethical analytics culture values experimentation, in which pre-registering⁴ success metrics and conducting A/B or uplift tests to foster transparency and continuous learning.

Managerial Insights

- Clarify what are the relevant decisions, constraints, and cost asymmetries before, during, and after modeling. Think of it not as a sequential but as a cyclical process.
- Validate with business-relevant metrics (precision, lift, cost, predicted outcomes).
- Communicate thresholds, not only model results. Quite often, it is useful to report multiple scenarios.
- Present an action plan with owners, Service Level Agreements, and KPI.
- Pilot, measure uplift, monitor drift, and iterate.

³ Sometimes C_{FP} and C_{FN} are known, on which case they should be used. Quite often, they are not known. In that case, you need to either make assumptions or ask the experts on the topic to make assumptions.

⁴ Behavioral researchers in business often preregister their studies in projects such as AsPredicted, from Wharton Credibility Lab (<https://credlab.wharton.upenn.edu/>). They do so to provide accountability for publication purposes and to avoid ethical concerns. Doing so would be unnecessary in business practice, but an internal pre-registration can promote ethical and professional reporting practices.

EXHIBIT 1

Table 1. Bridging Technical and Managerial Metrics

Technical	Plain Language	Typical Business Use
Precision = TP / (TP + FP)	How many flagged are truly positive?	Reduce wasted offers / false alarms
Recall = TP / (TP + FN)	How many true positives did we catch?	Safety/risk, churn catch rate
F1 = (2 P R) / (P + R)	Balance of precision vs. recall	Imbalanced settings
AUC and ROC	Rank-separation ability	Model selection (broad)
Lift @k%	Value concentration at top k%	Campaign sizing, staffing
MAPE / RMSE	Forecast error magnitude	Inventory, budgeting

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